Airbnb NYC Listings

Business Analysis Report

Prepared by: Md Mazhar

Dataset: Airbnb Open Data (NYC) 102,036 cleaned listings

Tools: Python (NumPy, Pandas), MySQL, Power BI

1. Executive Summary

This project analyzes 102,036 Airbnb listings across New York City, covering the complete analytics pipeline from raw data cleaning and SQL analysis to an interactive Power BI dashboard.
The analysis found that while Manhattan has the highest number of listings, Staten Island and Queens have substantially higher average reviews per listing, suggesting stronger historical guest engagement per property. However, availability data indicates that these may be established, long-standing listings with accumulated reviews rather than evidence of currently higher demand.
A key data-quality finding is that the **price variable shows no meaningful relationship with room type, borough, cancellation policy, or guest ratings**. In some cases, the relationship even contradicts real-world expectations, for example, 1-star listings being priced higher than 5-star listings. This indicates that the price data may be synthetically generated or unreliable. Consequently, **no business recommendations in this analysis rely on price as a genuine market signal**.

2. Business Questions

This project investigated the following ten business questions using SQL analysis on the cleaned dataset:

1. Which neighbourhood groups have the highest average prices?
2. Which room types generate the most listings?
3. Which areas have the highest review activity?
4. How does availability vary by neighbourhood?
5. Which hosts have the largest number of listings?
6. How does the cancellation policy relate to pricing?
7. What factors appear associated with higher prices?
8. Which neighbourhoods offer the best balance of price and reviews?
9. What percentage of listings are instant-bookable?
10. Which areas have the highest average availability?

3. Data Preparation

The raw dataset contained 102,599 rows and 26 columns, with a range of quality issues including missing values, incorrect data types, duplicate records, and logically impossible values. The cleaning process addressed several major categories of problems. Missing values in identifier and text-based columns, such as host_name and NAME, were filled with "Unknown" rather than dropped, since removing those rows would have discarded other valid information sitting in the same record. The last_review column was converted from text to a proper date type, and price and service_fee were converted from currency-formatted text to numeric values.

Invalid numerical entries were corrected based on their severity: 431 mildly negative availability_365 values were capped at 0, 2,752 mildly high values (366-411) were capped at 365, and one extreme outlier (3,677) was dropped rather than capped, since correcting a deviation that large would have meant fabricating a number rather than fixing a minor error. Similarly, 13 rows with impossible negative minimum_nights values were removed entirely, since there was no reliable way to infer what the correct value should have been. Additionally, 541 exact duplicate rows were identified and removed.

After cleaning, the dataset was reduced to 102,036 rows and 22 columns, with four low-value columns (country, country code, license, house_rules) dropped entirely. Every decision throughout this process followed a consistent approach: understand the cause of a problem before deciding how to fix it, and prioritize honest, defensible outcomes over technical convenience.

4. Analysis

Pandas Analysis

The most significant finding from the cleaning phase concerned the price column. An IQR outlier test returned zero outliers, and the price percentiles were suspiciously evenly spaced, a pattern rarely seen in real-world pricing data. This led to the conclusion that price was likely randomly generated rather than reflecting an actual market. A second notable finding was that host_id is nearly 1:1 with listings (102,035 unique hosts across 102,036 rows), directly contradicting the calculated_host_listings_count column, which claims some hosts have over 300 listings. Since host_id can be independently verified by counting rows and the other column cannot, host_id was treated as the more reliable source.

SQL Analysis

Business questions fell into two clear categories. Customer engagement metrics proved to be the most reliable and informative: Staten Island and Queens showed the highest average reviews per listing despite having far fewer total listings than Manhattan, which ranked lowest despite the highest listing volume. Cross-referencing this with availability data revealed that Staten Island also had the highest average availability, suggesting these are long-standing, established listings with strong historical engagement rather than currently high demand. By contrast, every test involving price against borough, room type, cancellation policy, instant-bookable status, and guest rating showed no meaningful or logically consistent relationship. Guest ratings even ran in the opposite direction expected, with 1-star listings priced higher on average than 5-star listings, reinforcing the conclusion that price should not be trusted as a basis for business recommendations in this dataset.

Power BI Analysis

Visualizing the data added value beyond the raw SQL numbers in two main ways. The geographic map made listing density visible at a glance, showing clear clustering around Manhattan and Brooklyn that a table of numbers alone doesn't convey as immediately. The side-by-side reviews-and-availability chart made the borough-level pattern easy to spot visually, showing both metrics moving together across all five boroughs rather than requiring a reader to compare two separate tables. Filtering the "highest availability by neighbourhood" chart to only include areas with 20 or more listings also demonstrated, visually, how a small sample size can distort a ranking: several neighbourhoods with only 1-2 listings had initially appeared at the top before this correction.

Figure 1. Listing density by neighbourhood, plotted from raw latitude/longitude coordinates.

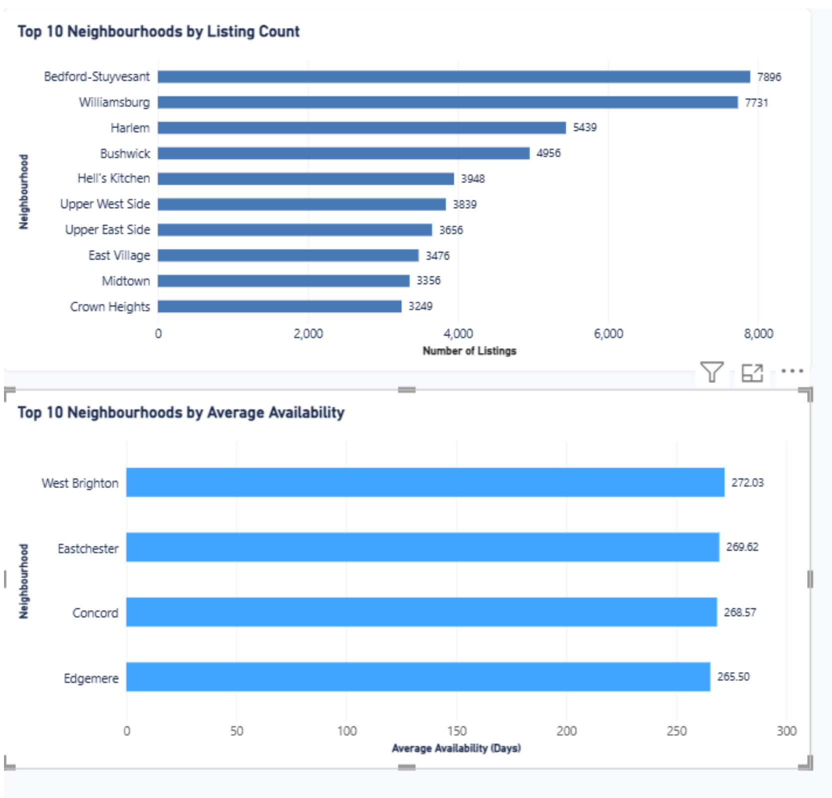

Figure 2. Top 10 neighbourhoods by listing count (top) and by average availability, filtered to areas with 20+ listings to avoid small-sample distortion (bottom).

5. Key Insights

- **Engagement is inversely tied to listing volume.** Staten Island and Queens show the highest average reviews per listing despite having far fewer listings than Manhattan, which ranks lowest, likely because less competition means each property captures more guest attention.
- **High reviews don't necessarily mean high current demand.** Staten Island also shows the highest average availability, suggesting these are established listings with strong historical engagement rather than currently in-demand properties.

Figure 3. Average reviews and average availability move together across boroughs, the pattern that revised the Staten Island "high demand" story into "established but currently quieter listings."

- **Price data shows no real-world signal.** An IQR test found zero outliers and suspiciously uniform percentiles; price showed no meaningful relationship with room type, borough, cancellation policy, or guest rating, and, in one case, ran backward (1-star listings priced higher than 5-star). This strongly suggests the price column is synthetically generated.

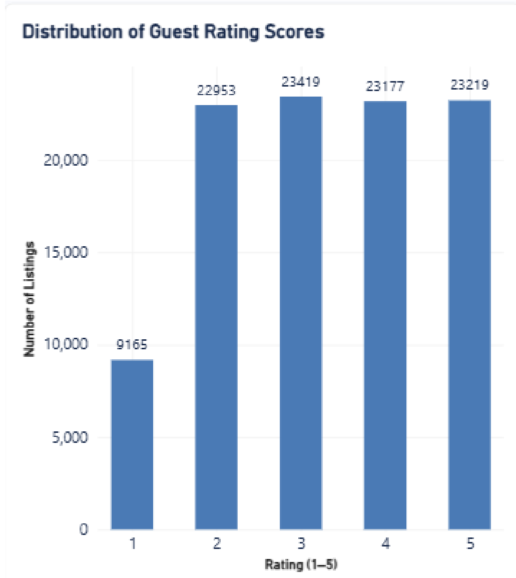

Figure 4. Distribution of guest rating scores, an unusually flat spread across 2–5 that further supports the synthetic-data concern.

- **Host concentration doesn't exist in this dataset.** host_id is nearly 1:1 with listings (102,035 unique hosts across 102,036 rows), directly contradicting the calculated_host_listings_count column, which claims some hosts have 300+ listings.

6. Recommendations

1. Investigate the gap between historical engagement and current demand in boroughs like Staten Island and Queens. These areas show strong review activity but also the highest availability, meaning listings are sitting unbooked more than in any other area, despite a positive guest history. Possible next steps include reviewing listing photos and descriptions for staleness, assessing increased competition from newer platforms, and evaluating whether targeted marketing could reconnect these well-reviewed listings with demand.
2. Audit the calculated_host_listings_count column before using it in any further analysis. This column contradicts a directly verifiable metric (host_id counts), and its calculation method is unknown. It should not be trusted or reported on until its source and derivation can be confirmed.
3. Replace or independently verify the price column before using it for any pricing-related analysis or decisions. Repeated testing found no realistic relationship between price and any other variable in the dataset, suggesting the values may be synthetically generated. Any future work involving pricing should be built on a verified, real-world price source rather than this column.

7. Limitations

- **Price does not behave like real-world data.** It shows no meaningful relationship with room type, borough, cancellation policy, or guest rating, and in the case of guest rating, runs in the opposite direction expected (lower-rated listings priced higher). Dozens of listings across different boroughs also share the exact same maximum price of $1,200.00, which is inconsistent with organic pricing. This strongly suggests the price was synthetically generated.

- **calculated_host_listings_count cannot be verified.** It contradicts a directly checkable metric (counting rows by host_id), and its calculation method is unknown, so it should not be relied on.
- **Small sample sizes can distort neighbourhood-level rankings.** For example, neighbourhoods with only 1-2 listings initially appeared at the top of an availability ranking simply due to limited data, not a genuine pattern; a minimum listing threshold was applied to correct this.
- **Several categorical fields show unusually even distributions.** instant_bookable, cancellation_policy, and host_identity_verified all split close to 50/50 (or evenly across categories), which is uncommon across so many unrelated fields simultaneously and adds further support to the concern that parts of this dataset may be artificially generated.
- **Review-related time analysis is limited by missingness.** last_review and reviews_per_month are missing for roughly 15,800 listings that have never been reviewed, an expected and meaningful pattern, not a data error, but it does mean any analysis of review trends over time only reflects a subset of listings.

8. Conclusion

This project demonstrates a complete, end-to-end data analysis workflow from raw data cleaning in Pandas, through structured analysis in SQL, to an interactive Power BI dashboard applied to a real-world-style dataset of over 100,000 Airbnb listings. The most reliable and actionable finding throughout was around guest engagement: boroughs with fewer listings, like Staten Island and Queens, showed meaningfully higher average reviews per listing than high-volume Manhattan, though further investigation revealed this reflects established, historically engaged listings rather than current demand.

Just as important as the findings themselves was the discipline of testing assumptions rather than accepting the data at face value. Repeated testing revealed that the price column does not behave like genuine market data, and that a column claiming to measure host listing concentration could not be trusted. Rather than building recommendations on top of unreliable numbers, this project treats identifying and documenting those limitations as a core part of the analysis itself, not a footnote to it.